A

T.Y. Mini Project Report

(Formative Assessment 2)

of

VSEC Course: Computational Tools for AIML

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Smart Vehicle Accident Detection and Alert System Using IoT and Machine Learning

By

Vivek Deshmukh **124B2E030**

Vipul Jadhav **124B2E038**

Dr. Ashwini S. Shinde
(Project Guide)

Department of Electronics & Telecommunication Engineering
Pimpri Chinchwad College of Engineering,
Savitribai Phule Pune University, Pune

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ABSTRACT

Road accidents are a major cause of fatalities worldwide, and delays in accident detection and emergency response often lead to increased loss of life. Traditional accident reporting systems rely on manual communication, which can be slow and unreliable, especially when victims are unable to seek help. This project presents an AI-based vehicle accident detection and monitoring system designed to improve road safety through automatic detection and real-time alert generation.

The proposed system integrates multiple sensors such as an accelerometer, gyroscope, temperature sensor, and microphone to continuously monitor vehicle conditions. A GPS module is used to track the real-time location of the vehicle. The collected data is processed using machine learning algorithms like Decision Tree and Random Forest to identify abnormal patterns indicating a possible accident. When an accident is detected, the system automatically sends an emergency alert along with the exact GPS location to predefined contacts or emergency services.

This approach reduces dependency on manual reporting and ensures faster response time, which is critical in saving lives. Additionally, the system can store accident-related data for further analysis and investigation. The proposed solution demonstrates how the integration of IoT and artificial intelligence can enhance vehicle safety and provide an efficient, reliable, and real-time accident detection system.

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[Information Regarding Sensors, Motors, meters, etc. & their uses.]

OR [Information of AI Algorithm]

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CHAPTER 1: INTRODUCTION

1. Introduction

Road accidents are a major concern across the world, leading to significant loss of life and property every year. One of the primary reasons for high fatality rates is the delay in accident detection and lack of immediate medical assistance. In many cases, victims are unable to inform emergency services due to severe injuries or unconsciousness, which further increases the risk.

Traditional accident reporting systems rely on manual communication methods such as phone calls or eyewitness reports. These approaches are often unreliable and time-consuming, especially in remote areas where help may not be readily available. Moreover, most vehicles do not have systems to continuously monitor important parameters like speed, acceleration, and location.

With the advancement of technologies such as Internet of Things (IoT), embedded systems, and machine learning, it has become possible to design intelligent systems capable of real-time monitoring and analysis. These systems can automatically detect accidents and send alerts without human intervention.

1.1. Motivation / Need of the Project

Road accidents occur frequently and often result in serious injuries or fatalities. A major issue in such situations is the delay in providing medical assistance, as victims may not be able to communicate their condition. This delay can significantly reduce the chances of survival.

Existing accident detection methods depend mainly on manual reporting, which is not always reliable or fast. In many cases, especially in isolated areas, accidents may go unnoticed for a long time. Additionally, there is a lack of systems that can continuously monitor vehicle conditions and detect abnormal events automatically.

The development of IoT and machine learning technologies provides an opportunity to solve this problem. By using sensors to monitor vehicle parameters and applying intelligent algorithms to analyze the data, accidents can be detected in real time.

Therefore, there is a need for an automated accident detection and alert system that can quickly identify accidents and send emergency notifications with accurate location details. This will help reduce response time, improve rescue operations, and save lives.

CHAPTER 2: LITERATURE SURVEY

2.1 Literature Survey Table

Sr. No.	Author(s) & Year	Title / Paper	Methodology / Approach	Findings / Results	Limitations / Gaps
1	Singh et al. (2022)	IoT-Based Vehicle Accident Detection	Accelerometer and GSM detect crash impact and send GPS alerts	Real-time accident notification	Fixed threshold detection reduces accuracy
2	Sharma & Patel (2021)	Smart Vehicle Black Box System	Embedded system logs speed, acceleration and GPS	Reliable crash evidence storage	No accident prediction capability
3	Kumar et al. (2023)	AI-Based Driver Behavior Monitoring	Machine learning analyzes IMU driving patterns	Detects risky driving behaviors	Needs large datasets
4	Zhang et al. (2022)	Edge AI Vehicle Safety System	Edge computing processes vehicle sensor data locally	Low latency decision making	Limited microcontroller power
5	Reddy et al. (2020)	Smartphone Accident Detection	Smartphone IMU detects sudden impact	Low-cost crash detection	Phone orientation affects accuracy
6	Chen et al. (2021)	Vehicle Event Data Recorder	Logs vehicle speed, braking and motion	Useful accident evidence	No predictive capability
7	Gupta et al. (2024)	TinyML Crash Detection System	TinyML model on microcontroller using accelerometer data	Fast and low-power inference	Limited memory
8	Lee et al. (2022)	IoT Accident Monitoring System	Sensors detect crash and send alerts to cloud	Quick emergency response	Network dependency

9	Wang et al. (2023)	Deep Learning Accident Prediction	Neural networks analyze driving datasets	High prediction accuracy	High computation requirements
10	Patel et al. (2024)	Vehicle Monitoring System	Camera, GPS and IMU sensors monitor vehicles	Improved monitoring	Higher system cost
11	Ahmed et al. (2020)	Accelerometer Crash Detection	IMU detects rapid acceleration changes	Accurate crash detection	False positives
12	Park et al. (2021)	Edge Vehicle Monitoring	Embedded system processes vehicle data locally	Reduced latency	Limited analytics capability
13	Brown et al. (2022)	Vehicle Telematics Analysis	Vehicle telemetry analyzed for patterns	Predictive safety insights	Privacy concerns
14	Silva et al. (2023)	Driver Risk Assessment	Deep learning classifies risky drivers	Improved safety	Requires GPU processing
15	Mehta & Shah (2020)	IoT Vehicle Black Box	Microcontroller records vehicle data	Low cost implementation	Limited processing power
16	Zhao et al. (2022)	Multi-Sensor Safety System	Accelerometer + Gyroscope + GPS sensors	Better crash detection	Needs sensor calibration
17	Li et al. (2021)	AI Driver Behavior System	AI analyzes aggressive driving patterns	Helps reduce accidents	Requires continuous data
18	Roy et al. (2023)	Embedded AI Monitoring	AI models integrated in embedded systems	Real-time abnormal driving detection	Resource limitations
19	Das et al. (2022)	Smart Road Safety System	IoT sensors monitor traffic	Improves safety awareness	Needs connectivity

20	Kim et al. (2023)	Edge Crash Detection	Edge AI analyzes vehicle motion data	Fast detection	Hardware constraints
21	Fernandez et al. (2021)	Vehicle Data Recorder	Records vehicle dynamics before crash	Useful investigation data	Storage limitations
22	Hassan et al. (2022)	Machine Learning Accident Detection	ML models classify crash events	Higher accuracy	Needs retraining
23	Kumar & Verma (2024)	Road Safety Monitoring	AI predicts accident risk	Early warning system	Dataset dependency
24	Johnson et al. (2020)	Automotive Black Box Review	Review of event data recorders	Highlights importance	Limited AI use
25	Gupta et al. (2024)	AI Crash Prediction System	AI predicts crash probability using sensor fusion	Proactive accident prevention	Further validation required

2.2 Summary of Literature Survey

From the literature survey, it is observed that many systems have been developed for accident detection using IoT, embedded systems, and machine learning techniques.

Early systems mainly used sensors like accelerometers and GSM modules to detect crashes and send alerts. These systems were simple but had limitations such as fixed threshold detection and lack of accuracy.

Later research introduced black box systems and event data recorders that store vehicle parameters for analysis. However, these systems do not provide real-time accident prediction.

Recent approaches use machine learning and deep learning techniques to analyze driving behavior and detect accidents more accurately. Edge computing and TinyML have also been used to process data locally with low latency.

Although these methods improve detection accuracy, they face challenges such as high computational requirements, need for large datasets, hardware limitations, and dependency on network connectivity.

2.3 Summary of Literature Survey

Based on the survey, the following gaps are identified:

- Most systems use fixed threshold detection, reducing accuracy
- Lack of real-time intelligent prediction
- Many systems require high computational power
- Dependency on internet/network connectivity
- Limited use of multi-sensor integration
- Some systems generate false positives
- No complete system combining IoT + ML + real-time alerting

These gaps highlight the need for an improved system.

2.4 Summary of Literature Survey

- **Problem Statement**
  Existing accident detection systems are either slow, inaccurate, or dependent on manual reporting. They lack real-time intelligent analysis and efficient alert mechanisms. There is a need for an automated system that can accurately detect accidents and send immediate alerts with location details.

- **Aim**
  To develop an AI-based vehicle accident detection and monitoring system using IoT sensors and machine learning for real-time accident detection and alert generation.

- **Objectives**
  - To preprocess and analyze sensor data
  - To develop a machine learning model for accident detection
  - To detect accidents automatically
  - To send emergency alerts with GPS location
  - To reduce response time and improve safety
  - To collect real-time data using sensors (IMU, GPS, etc.)

CHAPTER 3: METHODOLOGY

3.1 Project Outline

The proposed system is an AI-based vehicle accident detection and monitoring system that uses IoT sensors, embedded systems, and machine learning techniques.

The system continuously monitors vehicle parameters such as acceleration, motion, temperature, and sound using sensors. This data is collected by the ESP32 microcontroller and transmitted for processing. A machine learning model analyzes the sensor data to detect abnormal patterns that indicate an accident.

When an accident is detected, the system automatically sends an alert message along with the GPS location to emergency contacts or authorities. The system ensures real-time monitoring, fast detection, and quick response.

3.2 Block Diagram Explanation

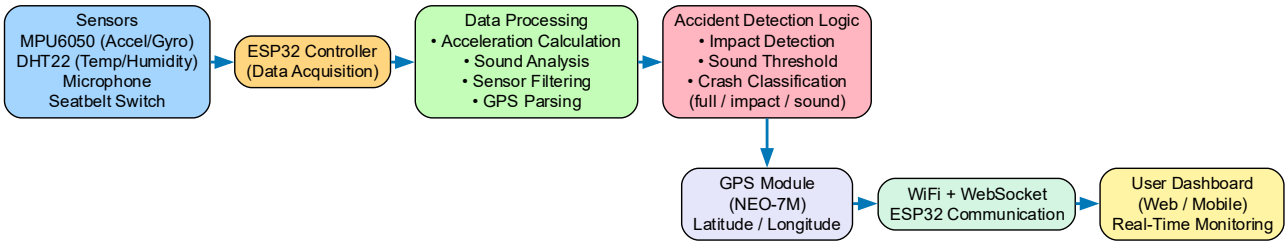


Figure 1: Proposed System Block Diagram

1. Sensors Block

Components:

- MPU6050 (Accelerometer + Gyroscope)
- DHT22 (Temperature & Humidity)
- Microphone (MAX4466)
- Seatbelt Switch

Function:

- Collects real-time data from the vehicle
- MPU6050 detects motion, tilt, and sudden impact
- Microphone detects crash sound
- DHT22 monitors environmental conditions
- Seatbelt sensor checks safety status

2. ESP32 Microcontroller Block

Function:

- Acts as the main controller of the system
- Reads data from all sensors
- Performs initial processing and calculations
- Controls communication (WiFi, WebSocket)
- Executes the accident detection logic

3. Data Processing Block

Function:

- Converts raw sensor data into useful values
- Calculates:
  - Acceleration (m/s^2)
  - Delta acceleration (removing gravity)
  - Sound amplitude
- Filters noise and invalid readings
- Prepares data for decision making

Ensures accurate and clean data

4. Accident Detection Block

Function:

- Detects accident using conditions:
  - High impact (acceleration threshold)
  - Loud sound (sound threshold)
- Classifies crash type:
  - Full accident
  - Impact only
  - Sound only
- Generates alert signal

Core decision-making block of system

5. GPS Module (NEO-7M)

Function:

- Provides real-time location (latitude, longitude)
- Tracks vehicle speed
- Helps in locating accident position

6. Communication Block (WiFi + WebSocket)

Function:

- Connects ESP32 to internet or hotspot
- Sends data wirelessly
- Uses WebSocket for real-time communication
- Transmits sensor data and accident alerts

7. Dashboard / Alert System

Function:

- Displays real-time data on web/mobile dashboard
- Shows:
    - Speed
    - Temperature
    - Location
    - Accident status
- Sends alert messages with GPS link

8. Random Forest Algorithm

Description:

- Combination of multiple decision trees
- Each tree gives a prediction
- Final result is based on majority voting

Working:

- Takes sensor data as input
- Multiple trees analyze data
- Output is more accurate compared to single tree

Advantages:

- High accuracy
- Reduces overfitting
- Handles complex data patterns

CHAPTER 4: DESIGN & IMPLEMENTATION OF PROPOSED SYSTEM

4.1 Circuit Diagram

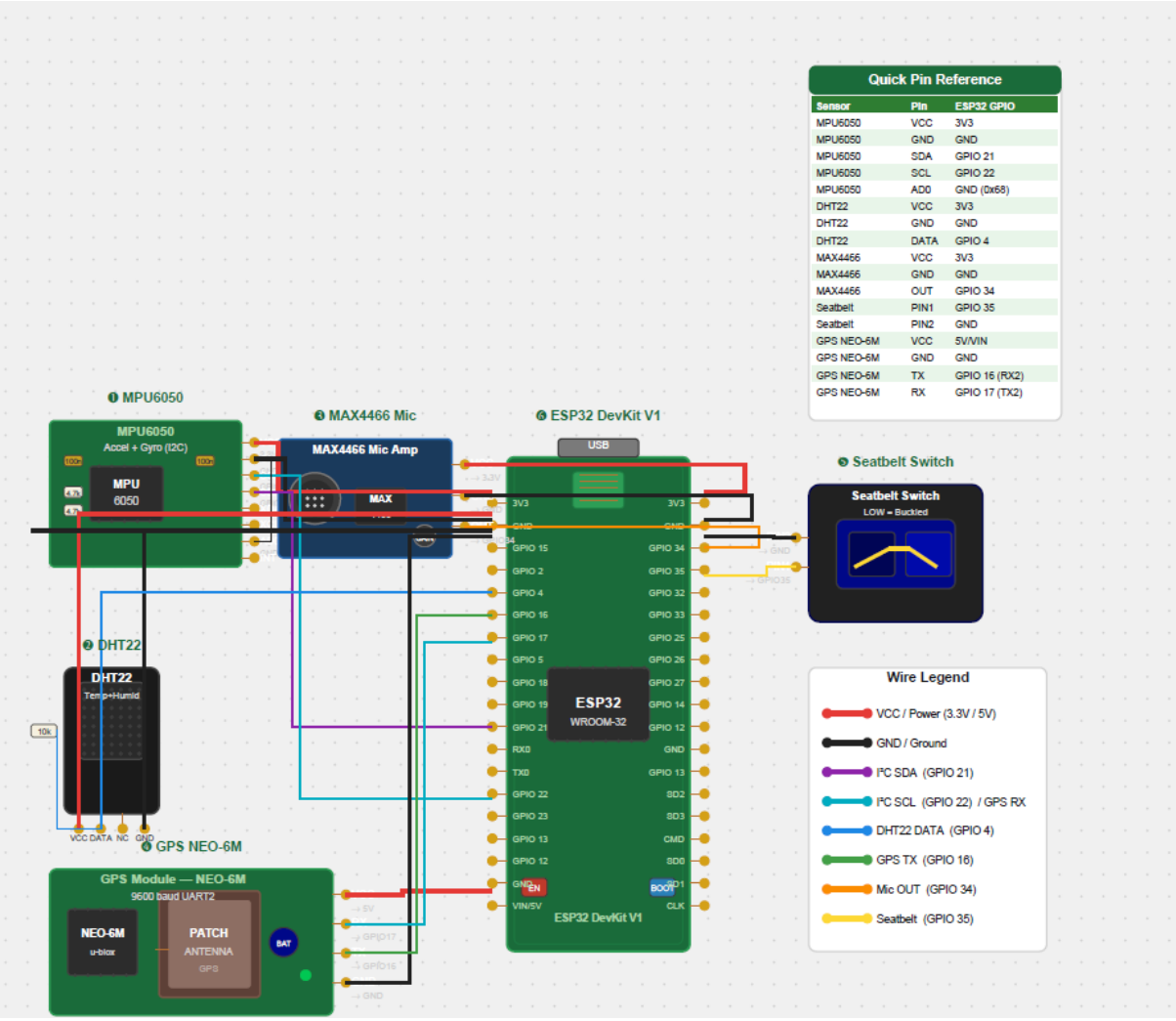


Figure 2: Proposed System Circuit Diagram

4.2 Hardware specification

1. ESP32 Microcontroller

- **Processor:** Dual-core Tensilica LX6
- **Operating Voltage:** 3.3V
- **Clock Speed:** Up to 240 MHz
- **Connectivity:** Wi-Fi and Bluetooth
- **GPIO Pins:** Multiple digital and analog pins

2. MPU6050 (Accelerometer + Gyroscope)

- **Type:** 6-axis IMU sensor
- **Voltage:** 3.3V
- **Interface:** I2C
- **Measurement:** Acceleration and angular velocity

3. DHT22 (Temperature & Humidity Sensor)

- **Voltage:** 3.3V–5V
- **Temperature Range:** -40°C to 80°C
- **Humidity Range:** 0–100%
- **Interface:** Digital

4. MAX4466 Microphone Module

- **Voltage:** 2.4V–5V
- **Output:** Analog signal
- **Gain:** Adjustable

5. GPS Module (NEO-7M)

- **Voltage:** 3.3V–5V
- **Interface:** UART (TX/RX)
- **Accuracy:** ~2.5 meters
- **Output:** Latitude, Longitude, Speed

6. Seatbelt Switch

- **Type:** Digital switch
- **Working:** HIGH/LOW signal

7. Power Supply

- **Voltage:** 3.3V / 5V regulated
- **Source:** Battery or adapter

8. Connecting Components

- Jumper wires
- Breadboard / PCB
- Resistors (if required)

4.3 Design Consideration

The design of the AI-based vehicle accident detection system is developed by considering various technical and practical factors to ensure reliability, accuracy, and real-time performance.

1. Real-Time Performance

The system must detect accidents instantly to reduce response time. Therefore, fast data acquisition and processing using ESP32 and efficient algorithms are considered.

2. Accuracy of Detection

To improve accuracy:

- Multiple sensors (IMU, microphone, GPS) are used
- Machine learning techniques are applied instead of fixed threshold methods
- Noise filtering and data preprocessing are included

3. Sensor Selection

Sensors are selected based on:

- High sensitivity (MPU6050 for motion detection)
- Reliability (DHT22, GPS module)
- Low cost and easy availability

4. Power Consumption

The system is designed to consume low power:

- ESP32 with efficient processing
- Optimized sensor usage
- Suitable for battery-operated systems

5. Communication Efficiency

WiFi and WebSocket communication are used for:

- Real-time data transmission
- Fast and reliable communication
- Easy integration with web dashboard

6. System Reliability

The system should work continuously without failure:

- Proper error handling
- Sensor calibration (gravity baseline in MPU6050)
- Stable connections

7. Scalability

The design allows future improvements:

- Integration with mobile applications
- Cloud storage
- Advanced AI models

8. Cost Effectiveness

Low-cost components like ESP32 and basic sensors are used to make the system affordable and practical.

9. Safety Considerations

- Proper wiring and insulation
- Stable power supply
- Secure data transmission

CHAPTER 5: SOFTWARE IMPLEMENTATION

5.1 Software Required

The following software tools are used for the implementation of the proposed system:

- **Arduino IDE**
  Used for programming the ESP32 microcontroller and interfacing sensors.
- **Python**
  Used for developing machine learning models and data analysis.
- **HTML, CSS, JavaScript**
  Used for designing the web-based dashboard for real-time monitoring.
- **Web Browser (Chrome/Edge)**
  Used to run and visualize the dashboard.

5.2 Software Specification

1. **Arduino IDE**

   - **Language:** Embedded C / C++
   - **Function:**
     - Reads sensor data (MPU6050, DHT22, microphone, GPS)
     - Performs real-time calculations (acceleration, sound)
     - Implements accident detection logic
     - Sends data via WiFi using WebSocket

2. **Python (Machine Learning)**

   - **Libraries Used:**
     - NumPy
     - Pandas
     - Scikit-learn
   - **Function:**
     - Data preprocessing
     - Feature extraction
     - Model training (Decision Tree, Random Forest)
     - Prediction of accident conditions

3. **Web Dashboard (HTML + JS)**

   - **Technologies Used:**
     - HTML (structure)
     - CSS (styling, UI design)
     - JavaScript (logic + real-time updates)
   - **Function:**
     - Receives real-time data via WebSocket
     - Displays:
       - Speed

        - Impact force
        - Temperature & humidity
        - GPS location
        - Seatbelt status
    - Shows alerts (Accident / Warning)
    - Provides event log and map link

5.3 Implementation

The system is implemented in three main stages:

1. Data Acquisition (Arduino / ESP32)

- Sensors collect real-time data:
    - MPU6050 → acceleration & motion
    - Microphone → sound amplitude
    - DHT22 → temperature & humidity
    - GPS → location & speed
- ESP32 processes data:
    - Calculates total acceleration
    - Removes gravity (delta acceleration)
    - Detects crash using thresholds
- Data is converted into JSON format and sent via WebSocket.

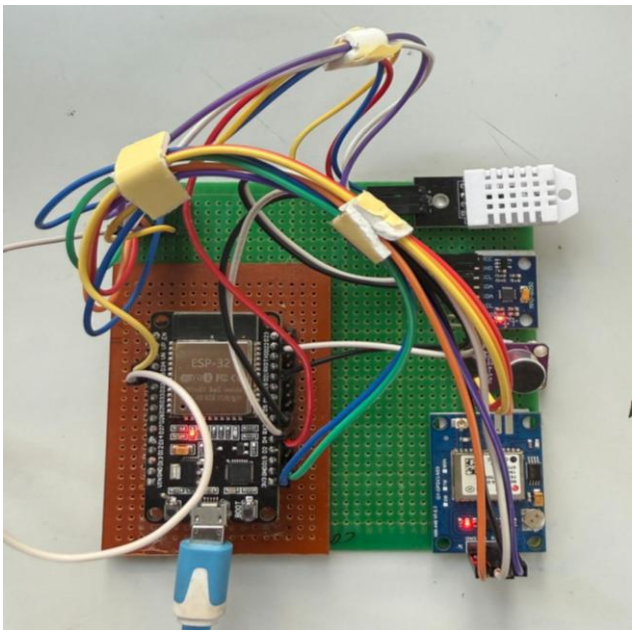
2. Machine Learning Model (Python)

- Sensor data is collected and stored as dataset
- Data preprocessing is performed:
    - Noise removal
    - Normalization
- Model training:
    - Decision Tree
    - Random Forest
- Model predicts:
    - Normal condition
    - Accident condition

3. Real-Time Dashboard (HTML Interface)

- Web dashboard connects to ESP32 using WebSocket (Port 81)
- Displays live data:
    - Speed, impact, sound, GPS
- Shows:
    - Normal driving
    - Hard impact
    - Accident detected
- Alerts:
    - Red warning screen
    - Crash notification
    - GPS link for location

- Event logging:
    - Stores all events
    - Option to download CSV



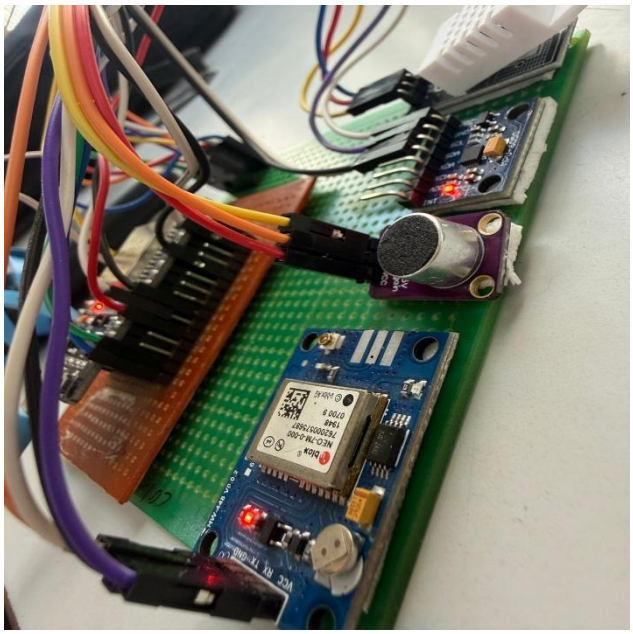


Figure 3: Proposed System Implementation

CHAPTER 6: TESTING/COMPARATIVE ANALYSIS

6.1 Testing

Testing is performed to verify the performance and reliability of the proposed accident detection system under different conditions.

Types of Testing Performed

- **Unit Testing**
  Each component (sensor, ESP32, GPS, dashboard) is tested individually.
- **Integration Testing**
  All components are tested together to ensure proper communication and data flow.
- **System Testing**
  Complete system is tested in real-time conditions.

Table 1: Proposed System Testing

Test Case	Input Condition	Expected Output	Result
Normal Driving	Low acceleration, low sound	No alert	Passed
Hard Impact	High acceleration	Impact detected	Passed
Crash Sound	High sound amplitude	Sound alert	Passed
Full Accident	High acceleration + sound	Accident detected	Passed
Seatbelt Off	Seatbelt unbuckled	Warning message	Passed
GPS Test	Valid signal	Correct location	Passed

6.2 Testing Strategies & Test Procedures

Testing Strategy

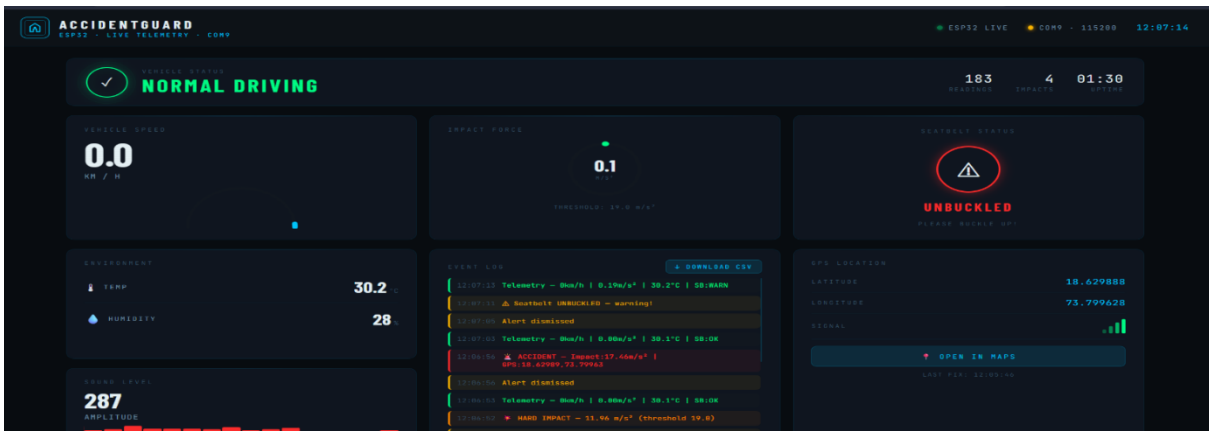
- Use **real-time sensor inputs**
- Simulate accident conditions manually
- Compare output with expected results
- Validate system response time

Test Procedure

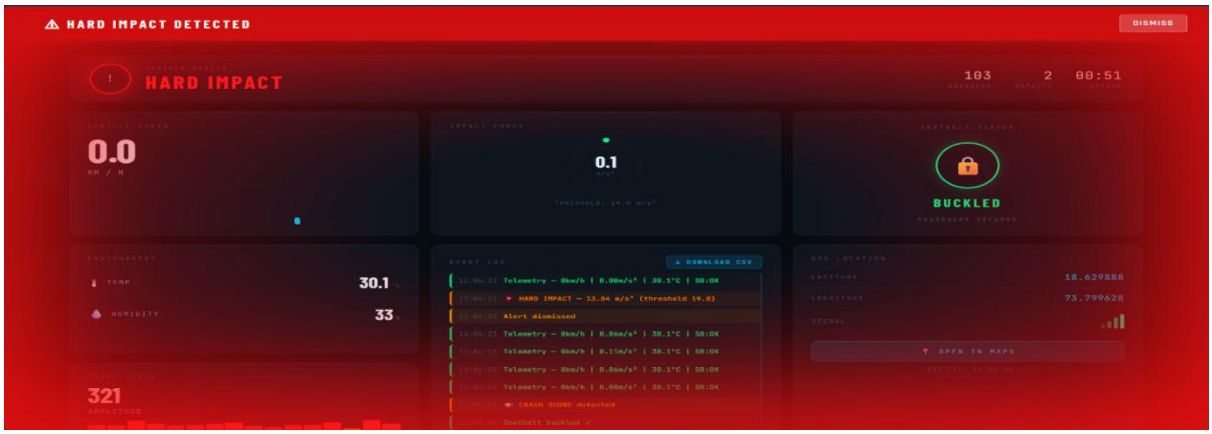
1. Connect all hardware components properly
2. Upload Arduino code to ESP32
3. Start WiFi and connect dashboard
4. Monitor sensor data on dashboard
5. Perform different test conditions:
    - Shake device → simulate impact
    - Clap or loud sound → simulate crash sound
    - Disconnect seatbelt → test warning
6. Observe system response and alerts
7. Record results for analysis

6.3 Result and Analysis

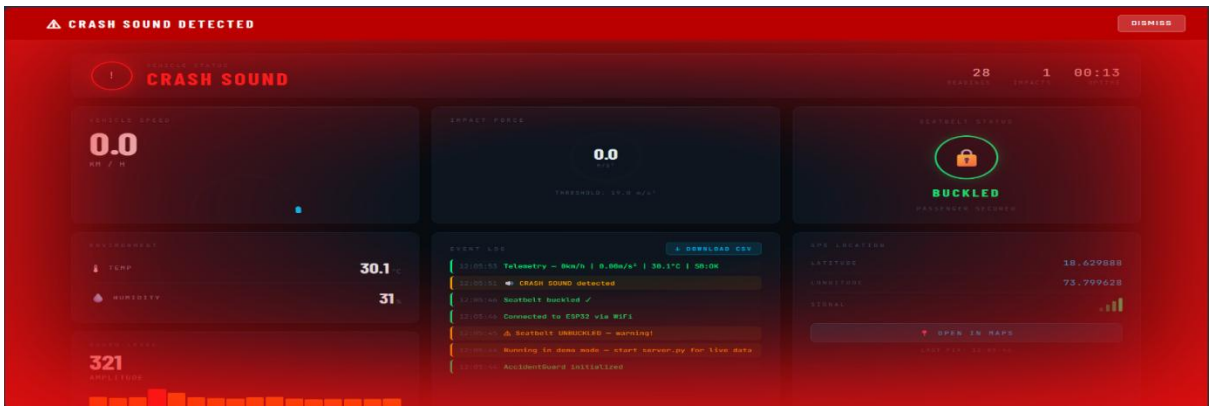
1. Normal Driving



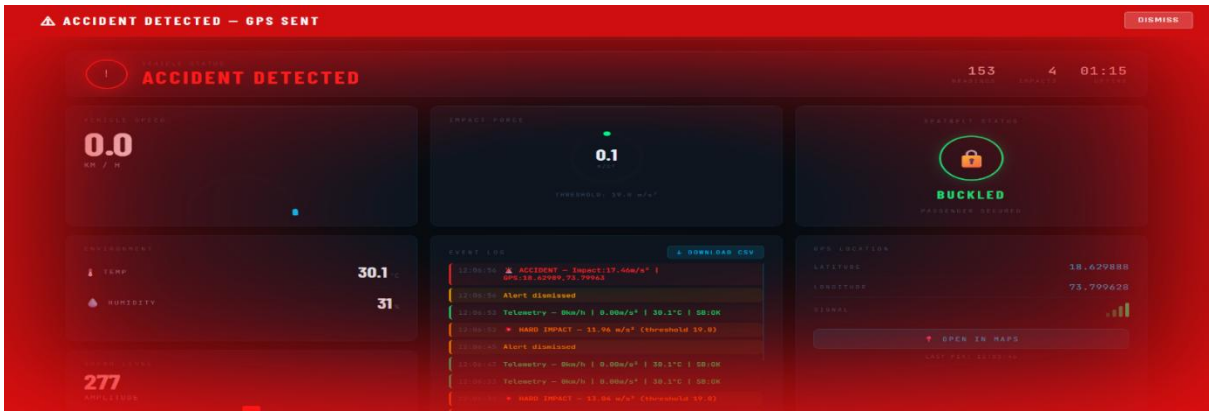
2. Hard Impact



3. Crash Sound



4. Accident Detected



Results

- System successfully detects:
  - Normal driving condition
  - Hard impact
  - Crash sound
  - Full accident
- Real-time data is displayed on dashboard
- GPS location is accurately transmitted
- Alerts are generated instantly

Performance Analysis

- **Accuracy:** High accuracy due to multi-sensor + ML approach
- **Response Time:** Fast (real-time detection within milliseconds)
- **Reliability:** Stable system performance
- **False Detection:** Reduced compared to threshold-only systems

Table 2: Comparative Analysis of Proposed System

Parameter	Traditional System	Proposed System
Detection Method	Threshold-based	ML + Sensor Fusion
Accuracy	Low	High
Response Time	Slow	Fast
Automation	Manual	Automatic
Real-Time Monitoring	Limited	Available
Alert System	Basic	GPS-based alerts

CHAPTER 7: ADVANTAGES & APPLICATIONS

7.1 Advantages

- **Real-Time Accident Detection**
  The system detects accidents instantly using sensor data and processing.

- **Fast Emergency Response**
  Automatically sends alerts with GPS location, reducing response time.

- **High Accuracy**
  Uses multiple sensors and machine learning for better detection accuracy.

- **Automatic Operation**
  Works without human intervention, unlike traditional systems.

- **Low Cost Implementation**
  Uses affordable components like ESP32 and basic sensors.

- **Wireless Communication**
  Uses WiFi and WebSocket for real-time data transmission.

- **User-Friendly Dashboard**
  Provides easy monitoring through a web interface.

- **Improved Safety**
  Helps in saving lives by quick detection and alert system.

7.2 Applications

- **Automotive Safety Systems**
  Used in cars and vehicles for accident detection.

- **Smart Transportation Systems**
  Can be integrated into intelligent traffic systems.

- **Emergency Services**
  Helps ambulance and rescue teams respond faster.

- **Fleet Management**
  Used by companies to monitor vehicles and driver safety.

- **Insurance Industry**
  Helps in accident analysis and claim verification.

- **Driver Monitoring Systems**
  Detects unsafe driving behavior.

- **Smart City Applications**
  Enhances safety in urban transportation systems.

7.3 Limitations

- **Dependence on Sensor Accuracy**
  Incorrect sensor readings may affect detection.

- **False Positives**
  Sudden jerks or loud sounds may trigger false alerts.

- **Network Dependency**
  Requires WiFi for real-time communication.

- **Limited Processing Power**
  ESP32 has limited capability for advanced AI models.

- **Environmental Factors**
  External noise or conditions may affect performance.

- **Initial Calibration Required**
  Sensors need proper calibration for accurate results.

CHAPTER 8: CONCLUSION & FUTURE SCOPE

8.1 Conclusion

The AI-based vehicle accident detection and monitoring system provides an effective solution for improving road safety and reducing emergency response time. The system integrates IoT sensors, ESP32 microcontroller, GPS module, and machine learning techniques to detect accidents in real time.

By continuously monitoring parameters such as acceleration, sound, and vehicle conditions, the system can accurately identify accident events. Once an accident is detected, it automatically sends alerts along with the exact GPS location to emergency contacts or authorities. This reduces dependency on manual reporting and ensures quicker medical assistance.

The implementation of a web-based dashboard further enhances the system by providing real-time monitoring and visualization of vehicle data. The combination of sensor fusion, intelligent processing, and wireless communication makes the system reliable and efficient.

Overall, the proposed system demonstrates how modern technologies like IoT and artificial intelligence can be used to enhance transportation safety and potentially save lives.

8.2 Future Scope

- **Integration with Mobile Application**
  A mobile app can be developed for better accessibility and real-time notifications.
- **Use of Advanced AI Models**
  Deep learning techniques can improve detection accuracy and prediction capabilities.
- **Cloud-Based Data Storage**
  Data can be stored and analyzed in the cloud for long-term insights and reports.
- **Integration with Emergency Services**
  Direct communication with hospitals and ambulance systems can be implemented.
- **Vehicle-to-Vehicle Communication (V2V)**
  Vehicles can share information to prevent accidents.
- **Improved Sensor Technology**
  More advanced sensors can be used for higher accuracy and reliability.
- **Automatic Vehicle Control**
  System can be extended to control vehicle functions in emergency situations.

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3. R. K. Sharma, "Accident Detection and Emergency Alert System Using Sensors," *International Journal of Electronics and Communication Engineering*, 2017.

4. Related Websites (Complete URL)

1. Python Official Website
   https://www.python.org
2. Arduino Official Website
   https://www.arduino.cc
3. ESP32 Documentation
   https://docs.espressif.com
4. Streamlit (for dashboards)
   https://streamlit.io
5. TinyGPS++ Library
   https://github.com/mikalhart/TinyGPSPlus
6. MPU6050 Documentation
   https://invensense.tdk.com/products/motion-tracking/6-axis/mpu-6050/

5. Related Books (Standard Format)

1. Russell, S. and Norvig, P., *Artificial Intelligence: A Modern Approach*, Pearson Education.
2. Haykin, S., *Neural Networks and Learning Machines*, Prentice Hall.
3. Kamal, R., *Embedded Systems: Architecture, Programming and Design*, McGraw Hill.